Climate CHANGE per interval − four ponds through the Holocene, survey plots in the Anthropocene

Holocene = UNCORRECTED CHELSA−TraCE21k (the series the analysis uses), 500−yr rolling mean across the 5 moving−window offsets, which are drawn faint behind.
Right of the dotted line: Anthropocene change across survey plots (present − past), one box per site. All paces are PER CENTURY at both ends.

Site (Anthropocene) MORA NOCA Pond (Holocene) JayBath LogWallow PantherPotholes ReflectionPond

Change over the interval (degC or mm)

MAT

degC per century

	min	mean	max
JayBath	−0.072	0.007	0.084
LogWallow	−0.073	0.007	0.084
PantherPotholes	−0.071	0.005	0.082
ReflectionPond	−0.068	0.007	0.087

MORA	0.952	2.474	4.249
NOCA	0.545	1.932	2.636

AP

mm per century

	min	mean	max
JayBath	−15.9	−2.196	12.8
LogWallow	−16.7	−2.362	13.2
PantherPotholes	−6.234	−0.436	4.823
ReflectionPond	−16.2	−2.395	12.6

MORA	−2401	−1570	−844
NOCA	−745	321	1359

MSpT

degC per century

	min	mean	max
JayBath	−0.110	0.007	0.117
LogWallow	−0.104	0.007	0.120
PantherPotholes	−0.128	0.005	0.111
ReflectionPond	−0.109	0.007	0.117

MORA	−1.802	1.570	5.091
NOCA	−2.701	0.393	2.139

SpP

mm per century

	min	mean	max
JayBath	−8.162	−0.252	5.217
LogWallow	−8.247	−0.272	5.485
PantherPotholes	−2.810	0.000	1.911
ReflectionPond	−7.991	−0.263	5.345

MORA	−496	−108	194
NOCA	255	493	807

MST

degC per century

	min	mean	max
JayBath	−0.091	−0.016	0.051
LogWallow	−0.092	−0.017	0.050
PantherPotholes	−0.107	−0.017	0.047
ReflectionPond	−0.103	−0.017	0.054

MORA	0.601	2.146	4.949
NOCA	1.331	2.198	2.635

SP

mm per century

	min	mean	max
JayBath	−3.894	−0.332	6.343
LogWallow	−4.113	−0.332	6.343
PantherPotholes	−1.962	−0.135	3.412
ReflectionPond	−3.937	−0.332	6.000

MORA	−292	−149	−72.0
NOCA	−108	157	456

MWT

degC per century

	min	mean	max
JayBath	−0.133	0.014	0.160
LogWallow	−0.137	0.015	0.161
PantherPotholes	−0.138	0.011	0.144
ReflectionPond	−0.129	0.015	0.153

MORA	0.453	2.907	4.505
NOCA	−0.741	1.250	2.388

WP

mm per century

	min	mean	max
JayBath	−8.644	0.236	9.001
LogWallow	−9.188	0.188	9.196
PantherPotholes	−2.723	0.320	3.399
ReflectionPond	−9.722	0.120	9.631

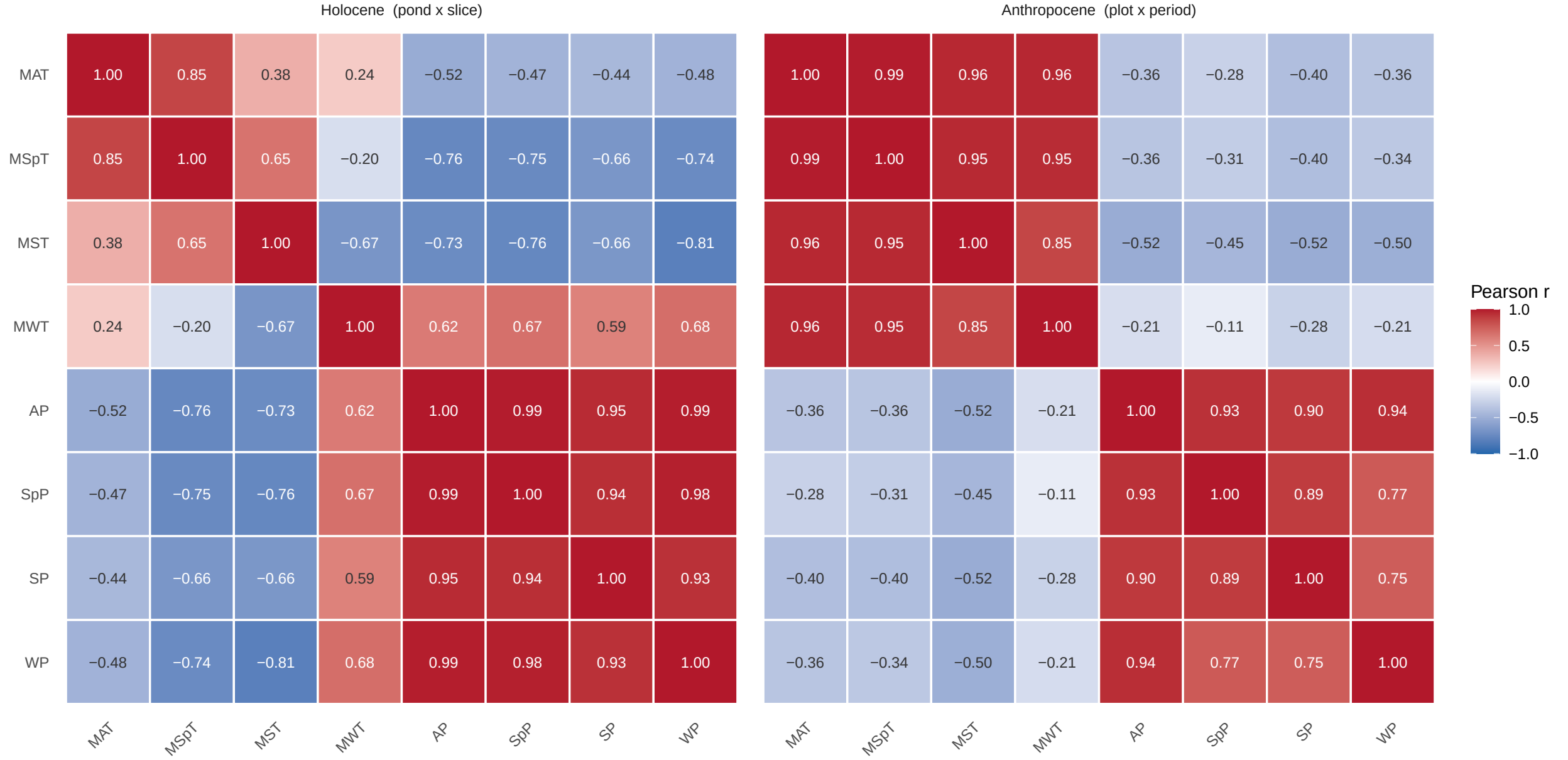
MORA	−1458	−874	−313
NOCA	−908	−403	−186

cal BP (time runs left to right) | Anthropocene

Holocene = the FULL TraCE21k record at each pond (all 111 slices, ~100 yr apart), NOT the ages where cores happen to have samples − so the line is the location's
climate history, independent of coring. Cores themselves reach JayBath 6841, LogWallow 6808, PantherPotholes 10609, ReflectionPond 6234 cal BP. Intervals differ (500 yr vs ~30 yr): these are CHANGES, not rates − see the

Correlation between climate variables, Holocene vs Anthropocene

The CCA learns the species−climate relationship from the ANTHROPOCENE plots and applies it to HOLOCENE samples.
Contrasting covariance structures can limit transfer; correlations alone do not test predictive validity.



Sign flips with |r| > 0.2 in both eras: MWT~MSpT (−0.20 vs 0.95); MWT~MST (−0.67 vs 0.85); MSpT~MWT (−0.20 vs 0.95); MST~MWT (−0.67 vs 0.85); AP~MWT (0.62 vs −0.21); SP~MWT (0.59 vs −0.28); WP~MWT (0.68 vs −0.21); MWT~AP (0.62 vs −0.21); MWT~SP (0.59 vs −0.28); MWT~WP (0.68 vs −0.21)